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High Performance Processors and Systems

June 9, 2020

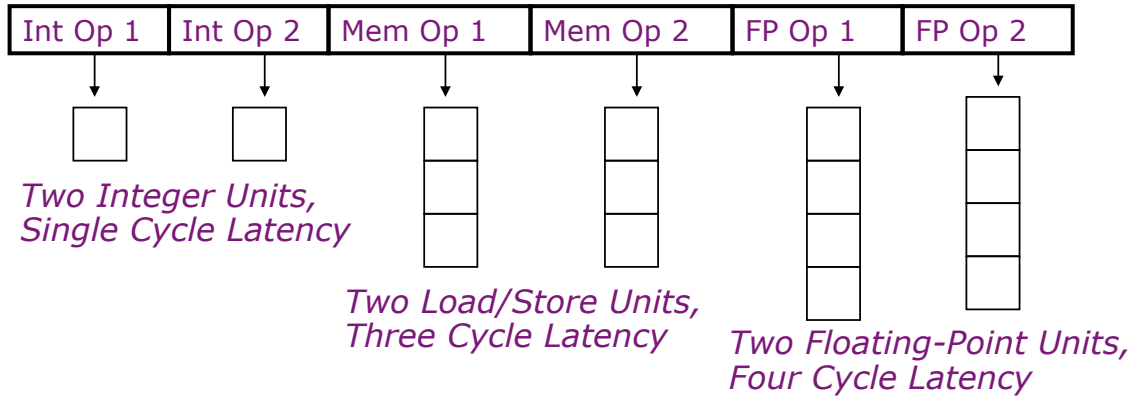
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Name
Last Name

Problem 1 (50%)	
Problem 2 (50%)	
Total (100%)	

Problem 1

Considering the following VLIW "architecture":



Considering the following portion of assembly, describe the corresponding VLIW code:

```
Loop:    LD F0, 0(R1)
         LD F6, 8(R1)
         LD F10, 16(R1)
         LD F14, 24(R1)
         FADD F4, F0, F2
         FADD F8, F6, F4
         FADD F12, F10, F14
         FADD F16, F14, F2
         SD F4, 0(R1)
         SD F8, 8(R1)
         SD F12, 16(R1)
         ADD R1, 32
         BNE R1, R2, Loop
         SD F16, -8(R1)
```

How many FLOPS/cycle?

Answer P1

Int1

Int 2

M1

M2

FP+

FPx

[illegible]

Problem 2

1. To implement cache coherency one of the protocols is the MESI concurrency protocol implemented for a write-invalidate write-back cache. Describe briefly the algorithm (also in term of graph) and complete the following table.

State	Cache up to date?	Memory up to date?	Others have a copy?	Cache can respond other's reads?
Modified				
Exclusive				
Shared				
Invalid				

2. Consider the following access pattern on a two-processor system with a direct-mapped, write-back cache with one cache block and a two cache block memory. Assume the MESI protocol is used, with write-back caches, write-allocate, and invalidation of other caches on write (instead of updating the value in the other caches).

Time	After Operation	P1 cache block state	P2 cache block state	Memory at block 0 up to date?	Memory at block 1 up to date?
0	P1: read block 1	Exclusive (1)	Invalid (0)	Yes	Yes
1	P2: read block 0				
2	P1: write block 0				
3	P2: read block 2				
4	P1: read block 1				
5	P2: write block 1				
6	P1: read block 1				
7	P2: write block 1				
8	P1: read block 0				
9	P2: write block 1				
10	P2: write block 1				
11	P1: read block 1				
11	P2: write block 1				
12	P1: read block 1				
13	P2: read block 1				